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Assignment 2

Case Study 1

Data for Questions 1 to 5

Clinic Follow-up Priority Scoring

A clinic uses a simple model to assign a follow-up priority score after remote consultation. The model does not m

x_1 : symptom persistence score, x_2 : abnormal reading score.

The model computes

$z = 3x_1 + 2x_2 - 2, \hat{p} = \sigma(z) = \frac{1}{1+e^{-z}}$

A record is sent for priority follow-up when $\hat{p} \geq 0.5$. For the sigmoid function, $\sigma(0) = 0.5$, positive z gives an output

1. Why is the sigmoid function suitable for this priority score?

- [x] It maps a real-valued input to a value between 0 and 1.
- [ ] It removes the need to compute any weighted input value.
- [ ] It always gives a hard class label before thresholding.
- [ ] It forces both input features to have identical weights.

2. For x_1 = 0.4 and x_2 = 0.6, what is the value of z?

- [ ] 0.2
- [ ] 0.6
- [x] 0.4
- [ ] 0.8

3. The rule $\hat{p} \geq 0.5$ is equivalent to which condition?

- [ ] $z < 0$
- [x] $z \geq 0$
- [ ] $z > 1$
- [ ] $z \leq -1$

4. Which input is not sent for priority follow-up?

- [ ] $x_1 = 0.8,\ x_2 = 0.1$
- [ ] $x_1 = 0.2,\ x_2 = 0.8$
- [ ] $x_1 = 0.5,\ x_2 = 0.4$
- [x] $x_1 = 0.1,\ x_2 = 0.4$

5. If the clinic wants fewer records to satisfy $\hat{p} \geq 0.5$, what should happen to the bias term -2?

- (selected) It should become more negative.
- It should be removed from the model entirely.
- It should be made equal to $x_1 + x_2$
- It should be increased to make every score larger.

Case Study 2
Data for Questions 6 to 10
Delivery Time Prediction Review Scoring
A logistics team trains a regression model to predict delivery delay in minutes. For four validation orders, the actual delay y and predicted delay $\hat{y}$ are:

Order	y	$\hat{y}$
1	12	10
2	20	24
3	15	13
4	18	21

The team uses squared error
$L = (y - \hat{y})^2, \quad \text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$
For gradient-based updates with respect to the prediction,
$\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y).$

6. What is the squared error for Order 2?

- 4
- (selected) 16
- 9
- 25

7. What is the mean squared error over the four orders?

- 7.25
- 8.00
- 9.50
- (selected) 8.25

8. Which order contributes the largest squared error?

- ( ) Order 1
- (•) Order 3
- ( ) Order 2
- ( ) Order 4

9. If $y = 20$ and $\hat{y} = 24$, what is $\frac{\partial L}{\partial \hat{y}}$?

- (•) 8
- ( ) −8
- ( ) 4
- ( ) −4

10. A second model predicts 11,21,14,19. Which model has lower MSE?

- ( ) The original model, because its MSE is 8.25.
- (•) The second model, because its MSE is 1.00.
- ( ) Both models, because their errors are identical.
- ( ) Neither model, because MSE cannot be computed here.

Case Study 3
Data for Questions 11 to 15
Machine Energy Use Prediction
A facilities team monitors configurable machines in a shared workshop where different operating choices lead to different electricity usage patterns. Before deploying a larger m
x_1 : duration score, x_2 : load score.
The team uses a small feedforward network:
$h_1 = \sigma(x_1 + x_2 - 1)$, $h_2 = \sigma(2x_1 - x_2)$,
$\hat{y} = \sigma(3h_1 + 2h_2 - 2)$.
The hidden layer transforms the original features before the final output is computed. This lets the team discuss the order of computation, the role of nonlinear hidden units, and

11. Which quantities are computed first in the feedforward direction?

- ( ) The final loss value
- ( ) The parameter update
- (•) The hidden units h_1 and h_2
- ( ) The target label chosen by the model

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12. For $x_1 = 1$ and $x_2 = 1$, what is the input to the sigmoid in h_1?

- (•) 1
- ( ) 0
- ( ) 2
- ( ) 3

13. For $x_1 = 1$ and $x_2 = 0$, what is the input to the sigmoid in h_2

- ( ) −2
- ( ) −1
- ( ) 1
- (•) 2

14. Why can sigmoid hidden units introduce nonlinearity?

- ( ) They copy inputs without changing any value.
- (•) They apply a nonlinear function to weighted input combinations.
- ( ) They remove all weights from the network.
- ( ) They force the output to be exactly linear for all possible inputs.

15. What is the final output range of $\hat{y}$?

- Any real number from $-\infty$ to ∞
- Only the two values −1 and 1
- A value between 0 and 1
- A value greater than 1 for all inputs

Case Study 4
Data for Questions 16 to 20
Warehouse Routing Time Analysis
A warehouse studies how a single routing coefficient affects the predic
$\hat{y} = wx$.
For one training example, $x = 3$ and $y = 15$. The squared error loss is
$L(w) = (15 - 3w)^2$.
The team studies this loss before using gradient descent on a larger tra

16. What is $L(w)$ when $w = 4$?

- 6
- 4
- 12
- 9

17. Which value of w gives zero loss?

- w = 4
- w = 5
- w = 6
- w = 3

18. What is $\frac{dL}{dw}$ at $w = 4$?

- −18
- −6
- 18.0
- 6

18. What is $\frac{dL}{dw}$ at $w = 4$?

- -18
- -6
- 18.0
- 6

19. If gradient descent uses $w \leftarrow w - \eta \frac{dL}{dw}$ with $\eta = 0.1$, what is the next value from $w = 4$?

- 3.2
- 4.6
- 5.8
- 2.2

20. What does the gradient indicate in this example?

- It gives the class label directly without a model.
- It removes the input feature from the prediction.
- It makes the squared error impossible to compute.
- It shows how the loss changes as w changes.